

Competitive Programming

Section 35: Math Mixed Contest

Competitive Programming Course

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Table of contents

1	How to approach a math mixed contest	3
2	A. Blocked grid paths	4
3	B. Coprime pairs	6
4	C. At least one event	7
5	Common mistakes	8
6	Contest checklist	9

1 How to approach a math mixed contest

Math problems rarely announce their technique. Use statement shape:

Signal	Likely tool
paths with right/down moves	combinations
avoid one forbidden object	subtract bad paths
gcd equals 1	prime factors and inclusion-exclusion
at least one event happens	complement probability
many values up to 10^6	sieve / SPF

2 A. Blocked grid paths

Paths from (1,1) to (H,W) use:

```
H-1 down moves  
W-1 right moves
```

so total paths:

```
C(H+W-2, H-1)
```

Paths through blocked cell (R,C):

```
ways start -> blocked  
* ways blocked -> finish
```

Subtract those bad paths.

Worked example:

```
H = 3, W = 4  
blocked = (2,2)
```

All paths need 2 down moves and 3 right moves:

```
C(5,2) = 10
```

Paths from (1,1) to (2,2) need one down and one right:

```
C(2,1) = 2
```

Paths from (2,2) to (3,4) need one down and two right:

```
C(3,1) = 3
```

Bad paths:

```
2 * 3 = 6
```

Good paths:

$$10 - 6 = 4$$

The multiplication is valid because every bad path has a unique prefix ending at the blocked cell and a unique suffix starting from it.

3 B. Coprime pairs

Two numbers are not coprime iff they share at least one prime factor.

When processing a new number x , factor it into distinct primes:

```
x = 2^a * 3^b * 5^c -> distinct primes [2,3,5]
```

For every nonempty subset of these primes, store how many previous numbers were divisible by their product. Inclusion-exclusion gives how many previous numbers share at least one prime factor with x .

```
previous coprime with x = previous_count - share_any_prime
```

Then add x to all subset-product counters.

This works because the subset products are square-free: each distinct prime is used at most once.

Example with previous numbers:

```
previous = [6, 10, 21]
x = 30 = 2 * 3 * 5
```

Numbers sharing a factor with 30:

```
divisible by 2: 6,10    -> 2
divisible by 3: 6,21    -> 2
divisible by 5: 10      -> 1
```

The sum $2+2+1$ counts 6 twice if it has factors 2 and 3, so subtract pair intersections:

```
divisible by 2*3: 6      -> 1
divisible by 2*5: 10     -> 1
divisible by 3*5: none   -> 0
```

Then add triple intersections:

```
divisible by 2*3*5: none -> 0
```

So:

```
share_any = 2 + 2 + 1 - 1 - 1 - 0 + 0 = 3
```

All three previous numbers are not coprime with 30, so there are 0 previous coprime pairs involving this x .

4 C. At least one event

For independent events:

$$P(\text{at least one happens}) = 1 - P(\text{none happen})$$

And:

$$P(\text{none happen}) = \text{product}_i (1 - p_i/q_i)$$

Use modular inverses for each q_i .

Example:

$$\begin{aligned} P(A) &= 1/2 \\ P(B) &= 1/3 \\ P(C) &= 1/4 \end{aligned}$$

Probability none happen:

$$(1/2) * (2/3) * (3/4) = 1/4$$

Probability at least one happens:

$$1 - 1/4 = 3/4$$

The events may overlap in the outcome space, so adding $1/2 + 1/3 + 1/4$ would be wrong.

5 Common mistakes

Subtracting paths through a blocked cell twice.

Counting prime powers instead of distinct primes in coprime-pair inclusion-exclusion.

Forgetting that value 1 has no prime factors and is coprime with all previous numbers.

Adding probabilities of events directly when they overlap.

6 Contest checklist

1. Translate the story into a known counting/probability shape.
2. Identify bad objects if subtraction or complement is easier.
3. Check whether values need preprocessing.
4. Verify edge cases: blocked start/end, value 1, probability 0 or 1.